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P07738 · PMGE_HUMAN

Protein[i]
Bisphosphoglycerate mutase

Gene[i]
BPGM

Status[i]
UniProtKB reviewed (Swiss-Prot)

Organism[i]
Homo sapiens (Human)

Amino acids
259 (go to sequence)

Protein existence[i]
Evidence at protein level

Annotation score[i]
5/5

Tools | Download | Add to basket | Add a publication | Entry feedback

Entry | Variant viewer | Feature viewer | Genomic coordinates | Publications | External links

Function[i]

Plays a major role in regulating hemoglobin oxygen affinity by controlling the levels of its allosteric effector 2,3-bisphosphoglycerate (2,3-BPG). Also exhibits mutase (EC 5.4.2.11) activity. 1 Publication

Catalytic activity[i]

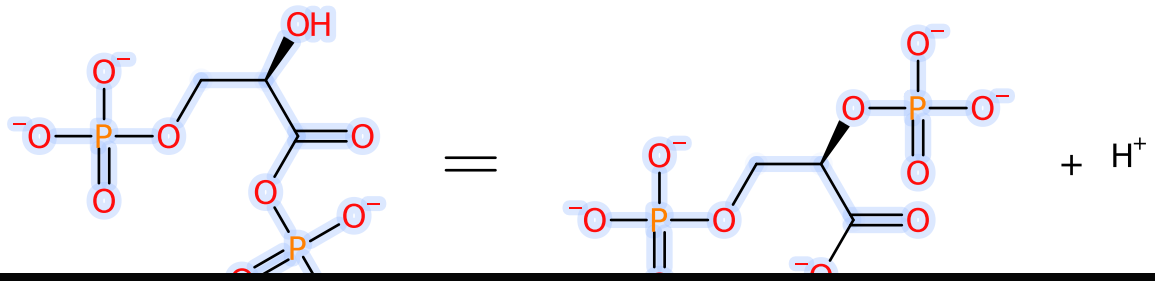
Rhea:17765

(2R)-3-phospho-glyceroyl phosphate = (2R)-2,3-bisphosphoglycerate + H^+ 1 Publication
EC:5.4.2.4 (**UniProtKB** | **ENZYME** | **Rhea**)

Hide reaction details

Reaction | Atom map

Color atoms ☑, Number atoms ☐, Confidence = 0.21419



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(2*R*)-3-phospho-glyceroyl phosphate » (2*R*)-2,3-bisphosphoglycerate » H^+ »

Rhea:15901

(2R)-2-phosphoglycerate = (2R)-3-phosphoglycerate 1 Publication

EC:5.4.2.11 (**UniProtKB** | **ENZYME** | **Rhea**)

View reaction details

Activity regulation[i]

At alkaline pH BPGM favors the synthase reaction; however, at lower pH the phosphatase reaction is dominant. Inhibited by citrate. 2 Publications

Features

Showing features for binding site[i], active site[i], site[i].

±	TYPE: All	ID	POSITION(S)	DESCRIPTION	
+	Binding site		10-17	substrate 1 Publication	Tools Add
+	Active site		11	Tele-phosphohistidine intermediate 1 Publication	
+	Binding site		23-24	substrate 1 Publication	
+	Site		29	Not glycated 1 Publication	
+	Site		46	Not glycated 1 Publication	
+	Binding site		62	substrate 1 Publication	
+	Active site		89	Proton donor/acceptor 1 Publication	
+	Binding site		89-92	substrate 1 Publication	Tools

Expand table

Gene Ontology[i]

GO annotations | GO-CAM models New

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Animals and fungi (Opisthokonta)



Cell color indicative of number of GO terms

ASPECT	TERM
Molecular Function	**bisphosphoglycerate mutase activity** Source:GO_Central
Molecular Function	**phosphoglycerate mutase activity** Source:UniProtKB-EC
Biological Process	**carbohydrate derivative catabolic process** Source:Reactome
Biological Process	**carbohydrate metabolic process** Source:ProtInc 1 Publication
Biological Process	**glycolytic process** Source:Ensembl
Biological Process	**respiratory gaseous exchange by respiratory system** Source:ProtInc 1 Publication

Access the complete set of GO annotations on QuickGO

PAN-GO

P07738 2 GO annotations based on evolutionary models

Keywords[i]

Molecular function

#Hydrolase

#Isomerase

Biological process

#Glycolysis

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PathwayCommons
P07738

Reactome
R-HSA-70171 Glycolysis

SABIO-RK
P07738

SignaLink
P07738

ENZYME
Search...

Names & Taxonomy

Protein names

Recommended name
Bisphosphoglycerate mutase

EC number
EC:5.4.2.4 (UniProtKB | ENZYME | Rhea) 1 Publication

Short name
BPGM

Alternative names
2,3-bisphosphoglycerate mutase, erythrocyte
2,3-bisphosphoglycerate synthase (EC:5.4.2.11 1 Publication) . EC:5.4.2.11 (UniProtKB | ENZYME | Rhea) 1 Publication
2,3-diphosphoglycerate mutase (DPGM)
BPG-dependent PGAM

Gene names

Name
BPGM

Organism names

Taxonomic identifier
9606 (NCBI)

Organism
Homo sapiens (Human)

Taxonomic lineage
Eukaryota > Metazoa > Chordata > Craniata > Vertebrata > Euteleostomi > Mammalia > Eutheria > Euarchontoglires > Primates > Haplorrhini > Catarrhini > Hominidae > Homo

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Secondary accessions

A4D1N9

Proteomes[i]

Identifier

UP000005640

Component[i]

Chromosome 7

Organism-specific databases

AGR

HGNC:1093

HGNC

HGNC:1093 BPGM

MIM

222800 phenotype

613896 gene

VEuPathDB

HostDB:ENSG00000172331

Subcellular Location[i]

UniProt Annotation | GO Annotation

cytosol Source:GO_Central

extracellular exosome Source:UniProtKB 1 Publication

Complete GO annotation on QuickGO

Disease & Variants[i]

Involvement in disease[i]

Erythrocytosis, familial, 8 (ECYT8)

3 Publications

Note

The disease is caused by variants affecting the gene represented in this entry

Description

An autosomal recessive disorder characterized by elevated serum hemoglobin and hematocrit, and

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See also
MIM:222800

Natural variants in ECYT8

VARIANT ID	POSITION(S)	CHANGE	DESCRIPTION
VAR_065367	62	R>Q	in ECYT8; dbSNP:rs751972865 1 Publication
VAR_065368	90	R>C	in ECYT8; mutation identified at protein level; marked decrease in synthase and mutase activities; no effect on phosphatase activity; dbSNP:rs121964925 2 Publications

Features

Showing features for natural variant[i].

±	TYPE	ID	POSITION(S)	DESCRIPTION
+	Natural variant	VAR_065367	62	in ECYT8; dbSNP:rs751972865 1 Publication
+	Natural variant	VAR_065368	90	in ECYT8; mutation identified at protein level; marked decrease in synthase and mutase activities; no effect on phosphatase activity; dbSNP:rs121964925 2 Publications

Keywords[i]

Disease
#Congenital erythrocytosis
#Disease variant
#Hereditary hemolytic anemia

Organism-specific databases

DisGeNET
669

MIM
222800 phenotype
613896 gene

MalaCards
BPGM

OpenTargets
ENSG00000172331

Orphanet
247378 Autosomal recessive secondary polycythemia not associated with VHL gene
714 Hemolytic anemia due to diphosphoglycerate mutase deficiency

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Pharos
P07738 Tbio

Chemistry

ChEMBL
CHEMBL5169112

Genetic variation databases

BioMuta
BPGM

ClinGen
HGNC:1093

DMDM
130350

GenCC
HGNC:1093

Expression[i]

Tissue specificity[i]

Expressed in red blood cells. Expressed in non-erythroid cells of the placenta; present in the syncytiotrophoblast layer of the placental villi at the feto-maternal interface (at protein level). 3 Publications

Gene expression databases

Bgee
ENSG00000172331 Expressed in secondary oocyte and 197 other cell types or tissues

ExpressionAtlas
P07738 baseline and differential

Organism-specific databases

HPA
ENSG00000172331 Tissue enhanced (bone)

Interaction[i]

Subunit[i]

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TYPE	ENTRY 1	ENTRY 2	NUMBER OF EXPERIMENTS	INTACT
BINARY	P07738	PGAM2 P15259	6	EBI-2115835, EBI-2511669

Protein-protein interaction databases

BioGRID
107137 30 interactors

IntAct
P07738 25 interactors

FunCoup
P07738 349 interactors

STRING
9606.ENSP00000376840

Chemistry

BindingDB
P07738

Miscellaneous

RNAct
P07738 protein

Structure[i]

3D structure databases

AlphaFoldDB
P07738

ModBase
Search...

SMR
P07738

PDBe-KB
Search...

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EvolutionaryTrace
P07738

Family & Domains

Sequence similarities

Belongs to the **phosphoglycerate mutase family**. **BPG-dependent PGAM subfamily**. Curated

Family and domain databases

- View the Phylogenomic databases for this entry within the **Homologs section**.
- View all family and domain features for this entry's canonical sequence in the **UniParc Feature Viewer**.

CDD
cd07067 HP_PGM_like 1 hit

FunFam
3.40.50.1240:FF:000012 Phosphoglycerate mutase 1 1 hit

Gene3D
3.40.50.1240 Phosphoglycerate mutase-like 1 hit

HAMAP
MF_01039 PGAM_GpmA 1 hit

InterPro
View protein in InterPro
IPR013078 His_Pase_superF_clade-1
IPR029033 His_PPase_superfam
IPR001345 PG/BPGM_mutase_AS
IPR005952 Phosphogly_mut1

NCBIfam
TIGR01258 pgm_1 1 hit
NF010713 PRK14115.1 1 hit

PANTHER
PTHR11931 PHOSPHOGLYCERATE MUTASE 1 hit

PIRSF
PIRSF000709 6PFK_2-Ptase 1 hit

PROSITE
View protein in PROSITE
PS00175 PG_MUTASE 1 hit

Pfam
View protein in Pfam
PF00300 His_Phos_1 2 hits

SMART
View protein in SMART
SM00855 PGAM 1 hit

SUPFAM
SSF53254 Phosphoglycerate mutase-like 1 hit

MobiDB
Search...

Sequence

Sequence status
Complete

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Length
259

Mass (Da)
30,005

Last updated
2007-01-23 v2

MD5 Checksum[i]
9F41027352BB0984DCB01ED20C949B36

MSKYKLIMLRHGEGAWNKENRFCSWVDQKLNSEGMEEARNCGKQLKALNFEFDLVFTSVLNRSIHTAWLILE
ELGQEWVPVESSWRLNERHYGALIGLNREQMALNHGEEQVRLWRRSYNVTPPPIEESHPYYQEIYNDRRYK
VCDVPLDQLPRSESLKDVLERLLPYWNERIAPEVLRGKTILISAHGNSSRALLKHLEGISDEDIINITLPTGVPILL
ELDENLRAVGPHQFLGDQEAIQAAIKKVEDQGKVKQAKK

Computationally mapped potential isoform sequences[i]

There is 1 potential isoform mapped to this entry

Align Add **View all**

Entry	Entry name	Gene name	Length
C9JH23	C9JH23_HUMAN	BPGM	78

Keywords[i]

Technical term
#3D-structure
#Direct protein sequencing
#Proteomics identification
#Reference proteome

Sequence databases

CCDS
CCDS5833.1

PIR
A31999 PMHUBM

RefSeq
NP_001280014.1 NM_001293085.2
NP_001432477.1 NM_001445548.1
NP_001715.1 NM_001724.5
NP_954655.1 NM_199186.3
XP_054214844.1 XM_054358869.1

NUCLEOTIDE SEQUENCE	PROTEIN SEQUENCE	MOLECULE TYPE	STATUS

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NUCLEOTIDE SEQUENCE	PROTEIN SEQUENCE	MOLECULE TYPE	STATUS
M23068 EMBL · GenBank · DDBJ	AAA51840.1 EMBL · GenBank · DDBJ	Genomic DNA	
M23067 EMBL · GenBank · DDBJ	AAA51840.1 EMBL · GenBank · DDBJ	Genomic DNA	
AK315439 EMBL · GenBank · DDBJ	BAG37827.1 EMBL · GenBank · DDBJ	mRNA	
CH236950 EMBL · GenBank · DDBJ	EAL24067.1 EMBL · GenBank · DDBJ	Genomic DNA	
CH471070 EMBL · GenBank · DDBJ	EAW83821.1 EMBL · GenBank · DDBJ	Genomic DNA	
BC017050 EMBL · GenBank · DDBJ	AAH17050.1 EMBL · GenBank · DDBJ	mRNA	

Genome annotation databases

Ensembl
ENST00000344924.8 ENSP00000342032.3 ENSG00000172331.13
ENST00000393132.2 ENSP00000376840.2 ENSG00000172331.13
ENST00000418040.5 ENSP00000399838.1 ENSG00000172331.13
ENST00000904928.1 ENSP00000574987.1 ENSG00000172331.13
ENST00000904929.1 ENSP00000574988.1 ENSG00000172331.13
More Ensembl links

GeneID
669

KEGG
hsa:669

MANE-Select
ENST00000344924.8 ENSP00000342032.3
NM_001724.5 NP_001715.1

UCSC
uc003vrv.4 human

Similar Proteins[i]

UniRef clusters[i]

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Homologs[i]

Orthologs & paralogs[i]

Phylogenomic databases

GeneTree
ENSGT00950000182926

HOGENOM
CLU_033323_1_1_1

InParanoid
P07738

OMA
TGWHDVP

OrthoDB
354304at2759

PAN-GO
P07738 2 GO annotations based on evolutionary models

PhylomeDB
P07738

eggNOG
KOG0235 Eukaryota

Organism-specific databases

AGR
HGNC:1093

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